

# TRYAMBAK KAUSHIK, PhD

## ENTERPRISE AI & PLATFORM LEADER | STRATEGY, DELIVERY, AND ENGINEERING

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### PROFILE

Enterprise AI and platform leader with 15+ years turning ambiguous business needs into production systems. At TELUS, shaped a multi-year platform strategy and delivery path and owned implementation of internal AI and data capabilities that influenced an executive build-versus-buy decision. Brings the judgment to set direction and the technical depth to challenge architecture, cost, risk, and delivery decisions.

### LEADERSHIP EVIDENCE

- Enterprise ownership: Set the implementation path and remained accountable from architecture through production operations for internal AI capabilities at TELUS.
- Executive influence: Produced enough internal capability and production evidence to inform C-suite decisions about a multi-million-dollar external platform.
- Organizational leverage: Established reusable infrastructure and analytics delivery patterns used directly or adapted by 4+ production teams within TELUS's governed data environment.
- Commercial judgment: Led proposal development and technical evaluation tied to CAD \$1M+ contracts, partnering with executive leadership through selection and award.

### PROFESSIONAL EXPERIENCE

#### Senior Technology Architect | TELUS, Canada | May 2023-Present

- Shaped a multi-year platform strategy and delivery path and owned engineering improvements across BiLayer, TELUS's analytics layer built on an existing 4-petabyte BigQuery Data Hub.
- Built and operated a governed natural-language interface using Gemini function calling on GKE with controlled access to BigQuery and Cloud SQL, designed for strict Canadian data-sovereignty requirements.
- Demonstrated a credible internal alternative to external software, influencing executive build-versus-buy direction while working within constrained funding and headcount.
- Introduced TELUS's first reusable Python Pulumi component pattern alongside the existing YAML path, encoding naming, IAM, security, and data-sovereignty controls for GCP resources. Documented and recommended by the platform team, the code was used directly or as the basis for components built by engineers across 4+ production teams.
- Established reusable dbt CI/CD and environment-specific deployment patterns for BiLayer, reducing manual setup and enabling teams to deliver governed analytics consistently through Superset and Looker.

#### Senior Data Engineer | Wavicle Data Solutions, Quebec, Canada | Aug 2022-Mar 2023

- Translated enterprise client requirements into automated data pipelines and scalable cloud analytics solutions, connecting architecture decisions to delivery constraints.

## EARLIER EXPERIENCE

### Senior Data Engineer | Syntax Cloud Inc., Ontario, Canada | Mar 2021-Jul 2022

- Architected cloud data platforms and led migration of on-premises systems to a governed data lake, coordinating delivery across ingestion, processing, orchestration, and infrastructure.
- Developed cloud billing analytics and identified cost-saving opportunities while incorporating client feedback through weekly delivery cycles.

### Engineer | Arcadis Canada Inc., Richmond Hill, Canada | Mar 2018-Mar 2021

- Coordinated multidisciplinary teams and stakeholders on complex infrastructure programs; led proposal development and technical bid evaluation for CAD \$1M+ in contracts with Vice President oversight through selection and award for First Nations communities.

### Senior Design Engineer | Infrastructure Technologies, Noida, India | Aug 2016-Mar 2018

- Led technical design, guided junior engineers, and coordinated client delivery for infrastructure projects.

### Technical Professional | Wood plc, Charlotte, USA | Feb 2015-Jun 2016

- Developed analytical models, technical designs, proposals, and project documentation for industrial facilities.

### Research Assistant | UNC Charlotte, USA | Jan 2014-Jan 2015

- Conducted research in structural health monitoring and sensor-based infrastructure analytics.

### Research Assistant | Michigan State University, USA | Aug 2009-Dec 2013

- Completed doctoral research on dynamic infrastructure systems; designed experiments, analyzed data, and delivered technical instruction and assessment.

### Assistant Manager | Ircon International Ltd., Kuala Lumpur, Malaysia | Jul 2007-Aug 2009

- Managed railway infrastructure operations, coordinating international teams, contractors, and delivery milestones.

## TECHNICAL CREDIBILITY

Hands-on depth in enterprise AI, distributed data platforms, cloud architecture, Kubernetes, production controls, and applied machine learning across GCP, AWS, and Azure. Built public demonstrations of governed AI-generated SQL execution and pre-deployment detection of software changes that create physical-system failures.

## PUBLIC WORK

SQL Execution Gate | Physical AI Release Gate | Historical reports on reinforcement learning and transformer-based recommendation systems - [tryambakkaushik.com](https://tryambakkaushik.com)

## CERTIFICATIONS EARNED

Google Professional Cloud Architect | AWS Certified Solutions Architect - Professional (SAP-C01) | AWS Certified Data Analytics - Specialty (DAS-C01) | Microsoft Azure Data Engineering (DP-203)

## EDUCATION

PhD, Civil Engineering, Michigan State University, 2014 | MS, Civil Engineering, Michigan State University, 2013  
MTech, Civil Engineering, Indian Institute of Technology Delhi, 2007 | BTech, Civil Engineering, National Institute of Technology Durgapur, 2005